Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 1 / 方法 1 [CO05-M01]

A recurrence is an exhaustive decomposition into disjoint smaller cases. Store exactly the information needed to determine legal continuations, state the empty and boundary cases, and explain why the update counts each object once. For a computed count, preserve the finite recurrence and an independently checkable certificate.

递推须把对象完备地分为互不重叠的较小情形。状态只保留决定合法后继所需的信息，明确空状态与边界，并解释每个对象为何恰计一次。依赖计算的计数应保留有限递推与可独立核验的证书。

Remember the suffix or last move needed by the rule

记住规则所需的后缀或末步

Procedure

1. Choose states recording the shortest suffix, last direction, used subset, or edge type that determines all legal extensions.

2. For each state list every legal next symbol or move and its new state.

3. Initialize the empty object explicitly, then add counts from disjoint predecessor states.

4. Sum only the terminal states satisfying the full endpoint or closure condition.

通用步骤

1. 状态记录决定后继所需的最短后缀、末方向、已用子集或末棱类型。

2. 对每态列出全部合法下一符号或动作及其新状态。

3. 明确空对象初值，再从互斥前驱状态相加。

4. 最后只汇总满足完整终点或闭合条件的终态。

> Example. Count binary words of length five avoiding the consecutive pattern 101.

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 1 / 方法 1 [CO05-M01] (continued / 续)

> 例题：求不含连续模式 101 的五位二进制串个数。

Worked solution

1. Use states $0, 1, 2$ for the longest suffix matching a proper prefix of 101: empty, 1, and 10.

2. The updates are $(a, b, c) \mapsto (a + c, a + b, b)$; from state 10, appending one is forbidden. Begin at $(1, 0, 0)$.

3. Lengths one through five give $(1, 1, 0), (1, 2, 1), (2, 3, 2), (4, 5, 3), (7, 9, 5)$. The answer is 21.

例题解析

1. 用状态零、一、二记录与 101 真前缀相同的最长后缀：空、一、十。

2. 更新为 $(a, b, c) \mapsto (a + c, a + b, b)$，状态十后不能再接一。初值 $(1, 0, 0)$。

3. 一至五位状态依次为 $(1, 1, 0), (1, 2, 1), (2, 3, 2), (4, 5, 3), (7, 9, 5)$，总数二十一。

A nonbacktracking walk must remember its previous edge; the current vertex alone is insufficient.

非回退游走须记住前一条棱，仅当前顶点不足以决定后继。

For a Hamilton path, use $H(S, v)$, with visited set S and last vertex v; append only unused adjacent vertices. A final cycle count needs the additional edge back to the start. For a cycle coloring, compare proper row colorings with equal versus unequal end colors; identifying equal ends yields $R_n + R_{n-1} = q(q-1)^{n-1}$.

Section 4.5 Recurrences, lattice paths & tilings

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Method 1 / 方法 1 [CO05-M01] (continued / 续)

哈密顿路径可用已访集合与末点状态 $H(S, v)$，只添加未访问相邻点；若计圈，还须检验最后能接回起点。环染色可按一行合法染色的两端相同或不同分类，合并同色端点得到 $R_n + R_{n-1} = q(q-1)^{n-1}$。

Practice / 配套练习: AIME-2001-I-Q14 (WB p. 677), AIME-2001-I-Q15 (WB p. 744), AIME-2005-I-Q13 (WB p. 679), AIME-2007-II-Q06 (WB p. 671), AIME-2008-I-Q11 (WB p. 681), AIME-2013-II-Q09 (WB p. 672), AIME-2015-II-Q12 (WB p. 682), AIME-2016-II-Q12 (WB p. 683), AIME-2021-II-Q08 (WB p. 788)

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 2 / 方法 2 [CO05-M02]

Extend acyclic paths by their accumulated totals

按累计总量扩展无环路径

Procedure

1. Use coordinates, total weight, or another quantity that strictly increases at every step.

2. Write a predecessor or forward-extension recurrence over that quantity, storing any running-value or turn information needed.

3. For a fixed multiset of moves, check the result independently with a multinomial interleaving count.

4. If turns are prescribed, replace maximal directional runs by positive compositions before summing.

通用步骤

1. 使用坐标、总重量等每步严格增加的量。

2. 按该量写前驱或向前扩展递推，并记录所需运行值或转弯信息。

3. 固定动作重数时，可用多项式交错计数独立核验。

4. 若规定转弯数，先把极大方向段转成正有序分拆，再求和。

> Example. A path uses steps $(1,0),(0,1),(2,1)$. Count paths from $(0,0)$ to $(3,2)$.

> 例题：路径允许步长 $(1,0),(0,1),(2,1)$，求从 $(0,0)$ 到 $(3,2)$ 的路径数。

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 2 / 方法 2 [CO05-M02] (continued / 续)

Worked solution

1. Let $F(x,y)=F(x-1,y)+F(x,y-1)+F(x-2,y-1)$, with $F(0,0)=1$ and out-of-range states zero.

2. Alternatively, without a long step there are three horizontal and two vertical moves, giving $\binom{5}{2}=10$.

3. With one long step, one move of each of the three types remains, giving $3!=6$; two long steps exceed the horizontal total. Thus $F(3,2)=16$.

例题解析

1. 令 $F(x,y)=F(x-1,y)+F(x,y-1)+F(x-2,y-1)$，初值 $F(0,0)=1$，范围外状态为零。

2. 独立核验：无长步时有三横两竖，排列数为 $\binom{5}{2}=10$。

3. 恰一长步时三类动作各一次，有 $3!=6$ 种；两长步横总量过大。因此共有十六条。

A recurrence ordered by total weight remains finite even when its running value can be negative.

即使运行值可为负，只要按严格增加的总重量排序，递推仍然有限。

For a nonstandard arithmetic operation, track both ordinary accumulated total s and running result v. A move of positive weight t advances to total $s+t$, so no cycle is possible in the dynamic program. A parity invariant may simplify the transition, but does not replace counting every admissible weight choice.

非标准运算可同时记录普通累计总量 s 与运行结果 v。新增正重量 t 后总量变为 $s+t$，故动态规划不会循环。奇偶不变量可简化转移，但不能替代对每个合法重量选择的计数。

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 2 / 方法 2 [CO05-M02] (continued / 续)

Practice / 配套练习: AIME-2005-I-Q13 (WB p. 679), AIME-2007-I-Q06 (WB p. 670), AIME-2018-II-Q08 (WB p. 674), AIME-2024-I-Q06 (WB p. 675), AIME-2026-II-Q14 (WB p. 689)

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 3 / 方法 3 [CO05-M03]

Insert and delete an extreme object to obtain a product recurrence

插删极值对象得到乘法递推

Procedure

1. Identify where the largest label must occur in a legal object.

2. Show deleting it preserves every local condition.

3. Count its legal reinsertion slots in every smaller object and use the resulting recurrence with its smallest valid initial case.

通用步骤

1. 确定合法对象中最大标签可出现的位置。

2. 证明删去它仍保持每个局部条件。

3. 对每个较小对象计算重新插入的合法位置数，结合最小有效初值递推。

> Example. An increasing rooted binary tree is required to be a single path and has labels $1, 2, 3, 4, 5$, increasing away from the root. Count its left-right shapes.

> 例题：递增二叉树只允许形成一条从根出发的路径，标签一至五沿路径递增，求左右子节点形状数。

Worked solution

1. The labels along the path are forced to be $1, 2, 3, 4, 5$.

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 3 / 方法 3 [CO05-M03] (continued / 续)

2. Insert the next largest label as the left or right child of the current final leaf. Deleting that largest leaf uniquely reverses this operation.

3. Four insertions each have two choices, so there are $2^4 = 16$ shapes.

例题解析

1. 路径上标签顺序被迫为一、二、三、四、五。

2. 每次把新增最大标签接为当前末叶的左或右子节点，删除最大叶可唯一逆转。

3. 四次插入各两种选择，共十六种形状。

If the deletion joins two former neighbors, check their new adjacency explicitly.

若删除后两位旧邻居相接，须明确检验新相邻关系。

For a permutation whose downward jumps are at most two, the new maximum may go at the end or immediately before either of the next two largest labels. Deleting it is safe because the newly adjacent predecessor is at most the old maximum minus one and its successor at least the old maximum minus two. This proves the three-slot recurrence rather than merely guessing it.

若排列每次下降至多二，新增最大值只能放末尾，或紧靠在接下来两个最大标签之前。删除它后，新的前项至多为旧最大值减一、后项至少为旧最大值减二，所以新邻接仍合法，由此严格得到三位置递推。

Practice / 配套练习: AIME-2006-I-Q11 (WB p. 680), AIME-2015-II-Q10 (WB p. 673)

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 4 / 方法 4 [CO05-M04]

Split at the earliest closed prefix or unique defect

按最早闭合前缀或唯一缺陷分解

Procedure

1. Choose a distinguished split that every object has uniquely.

2. Count the first component and the independent remainder.

3. Write a convolution recurrence, subtracting earlier component sizes when solving for the indecomposable count.

4. For a unique adjacent defect, forbid its neighboring extra defects before multiplying the two side counts.

通用步骤

1. 选择每个对象唯一具有的特殊分界。

2. 分别计算第一部件与独立余部。

3. 写卷积递推，求不可分计数时扣除更早部件大小。

4. 唯一相邻缺陷出现后，先禁止它两旁新增缺陷，再相乘左右计数。

> Example. Let $c_1, \ldots, c_5 = 1, 1, 3, 13, 71$ count indecomposable permutations, where no proper prefix is an initial value set. How many permutations of six labels have exactly two such consecutive components?

> 例题：不可分排列指没有真前缀恰为初始值集合，已知一至五项计数为 $1, 1, 3, 13, 71$。六标签排列中，恰有两个这样的连续部件者有多少？

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Method 4 / 方法 4 [CO05-M04] (continued / 续)

Worked solution

1. The first component has a unique size k, from one to five. Its values must be $1, \ldots, k$ and the second uses the others.

2. Each size contributes $c_k c_{6-k}$, with the second component standardized without changing its order.

3. The total is $2(1 \cdot 71) + 2(1 \cdot 13) + 3^2 = 177$. The earliest closed prefix makes the split unique.

例题解析

1. 第一部件大小 k 唯一，范围一至五，其值必须为一至 k，第二部件使用其余值。

2. 每个大小贡献 $c_k c_{6-k}$，第二部件标准化不改变相对顺序。

3. 总数为 $2(1 \cdot 71) + 2(1 \cdot 13) + 3^2 = 177$。最早闭合前缀保证分界唯一。

The word earliest or unique is a proof device that prevents a convolution from overcounting.

“最早”或“唯一”是防止卷积重复计数的关键。

An arbitrary permutation splits at its earliest closed prefix, giving $n! = \sum_{k=1}^{n} c_k (n-k)!$. For independent sets in a line, the choose-or-omit last position recurrence is $T_n = T_{n-1} + T_{n-2}$, with $T_0 = 1, T_1 = 2$. A prescribed unique adjacent pair leaves two independent no-adjacency blocks after excluding the two positions immediately beside it.

任意排列按最早闭合前缀分解，得到 $n! = \sum_{k=1}^{n} c_k (n-k)!$。直线上的不相邻子集按末位选或不选满足 $T_n = T_{n-1} + T_{n-2}$，初值 $T_0 = 1, T_1 = 2$。指定唯一相邻对后，排除其两侧紧邻位置，余下为两个独立不相邻区块。

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Method 4 / 方法 4 [CO05-M04] (continued / 续)

Practice / 配套练习: AIME-2018-II-Q11 (WB p. 685), AIME-2023-I-Q11 (WB p. 687)

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递推、格路与铺砌

Method 5 / 方法 5 [CO05-M05]

Reverse a deterministic process into exact first-hit levels

反转确定性过程得到首次命中层

Procedure

1. Define whether the hitting index counts terms or moves.

2. List every predecessor of a state from the original forward rule, including exceptional absorbing states.

3. Start at the target and generate one reverse level at a time, proving predecessor classes do not create duplicates or earlier hits.

4. For a queue or stack, reverse the full state or track original positions rather than guessing final labels.

通用步骤

1. 明确命中下标计算项数还是移动次数。

2. 从原正向规则列全每态前驱，包括特殊吸收态。

3. 从目标逐层反向生成，证明各前驱类不重复也不提前命中。

4. 队列或堆叠应逆转完整状态或追踪原位置，不能猜测最终标签。

> Example. Define $f(1)=1$; for even n, let $f(n)=n/2$; for odd $n>1$, let $f(n)=n-1$. Which starts reach one for the first time on their fourth term?

> 例题：定义 $f(1)=1$，偶数除以二，奇数且大于一时减一。哪些起点在第四项首次到一？

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Method 5 / 方法 5 [CO05-M05] (continued / 续)

Worked solution

1. The first level is $T_1 = \{1\}$ and the second is $T_2 = \{2\}$; the absorbing predecessor one must be excluded.

2. An even target z has predecessors $2z$ and $z+1$; an odd target greater than one has only $2z$. Thus $T_3 = \{3, 4\}$.

3. Reversing once more gives $T_4 = \{6, 8, 5\}$. Direct trajectories verify all three first-hit lengths.

例题解析

1. 初层为 $T_1 = \{1\}$、次层为 $T_2 = \{2\}$，须排除吸收前驱一。

2. 偶目标 z 有前驱 $2z, z+1$，大于一的奇目标只有 $2z$，所以第三层为 $\{3, 4\}$。

3. 再反向一步得到 $T_4 = \{6, 8, 5\}$，逐一正向检查均在第四项首次到一。

Different reverse branches can merge only if the forward map sends one source to two outputs, which a deterministic map cannot do.

同一反向层不同目标的前驱不会重复，因为确定性映射不可能把一个起点送到两个输出。

Practice / 配套练习: AIME-2000-I-Q15 (WB p. 676), AIME-2004-I-Q15 (WB p. 678)

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递推、格路与铺砌

Method 6 / 方法 6 [CO05-M06]

Normalize divisor chains to smaller integer states

将整除链归一到较小整数状态

Procedure

1. Choose an initial term or a final divisor so the remainder has a smaller integer parameter.

2. Divide out the common scale and translate strict growth into the new first-term restriction.

3. Sum only legal divisors, give the one-term case separately, and memoize by the strictly decreasing parameter.

通用步骤

1. 选择首项或末端约数，使余部具有较小整数参数。

2. 除去公共比例，把严格增长转成新首项限制。

3. 仅对合法约数求和，单独给单项情况，并按严格减小的参数缓存。

> Example. Count chains $1 = a_0 < a_1 < \cdots < a_t = 12$ in which every term divides the next.

> 例题：求从一严格递增到十二、且每项整除下一项的链数。

Worked solution

1. Let $D(n)$ count such chains ending at n, with $D(1) = 1$. The penultimate term is a unique proper divisor d of n, so $D(n) = \sum_{d \mid n, d < n} D(d)$.

2. This gives $D(2) = D(3) = 1, D(4) = 2, D(6) = 3$.

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Method 6 / 方法 6 [CO05-M06] (continued / 续)

3. Therefore $D(12) = D(1) + D(2) + D(3) + D(4) + D(6) = 8$. Each chain is counted by its unique penultimate term.

例题解析

1. 令 $D(n)$ 计终点为 n 的链，$D(1) = 1$。倒数第二项是唯一真约数 d，故 $D(n) = \sum_{d \mid n, d < n} D(d)$。

2. 依次得 $D(2) = D(3) = 1, D(4) = 2, D(6) = 3$。

3. 所以 $D(12) = 1 + 1 + 1 + 2 + 3 = 8$，每链按唯一倒数第二项计一次。

Dividing the remaining terms also changes their sum; subtract the removed normalized first term.

将余项除以首项会改变总和，还须扣除已移去的归一首项。

For a strictly increasing positive divisor chain of total s and first term at least two, define $g(s)$. The one-term chain contributes one. If its first term is $d \mid s$, $2 \le d < s$, dividing later terms by d leaves sum $s/d - 1$ and first term at least two. Hence $g(s) = 1 + \sum_{2 \le d < s, d \mid s} g(s/d - 1)$, with impossible small states zero.

对总和 s、首项至少二的严格递增正整数整除链，令计数为 $g(s)$。单项链贡献一；若首项为 $d \mid s$ 且 $2 \le d < s$，后续项除以 d 后总和为 $s/d - 1$，首项仍至少二。因此 $g(s) = 1 + \sum_{2 \le d < s, d \mid s} g(s/d - 1)$，不可行小状态设零。

Practice / 配套练习: AIME-2019-II-Q12 (WB p. 686)

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 7 / 方法 7 [CO05-M07]

Separate active and absorbing states in a weighted recurrence

在加权递推中区分活动与吸收状态

Procedure

1. Choose states with the same future transition law; verify any symmetry-based merging.

2. For counts, sum legal predecessor counts; for probabilities, multiply by transition probabilities.

3. Remove paths as soon as they hit the absorbing event, so later steps cannot count them again.

4. For expected remaining time, use $E = 1 + \sum pE_{next}$ with zero at absorbing states.

通用步骤

1. 选择未来转移规律相同的状态，按对称合并前须验证。

2. 计路径时相加合法前驱数，计概率时乘转移概率。

3. 一旦命中吸收事件便移除路径，避免以后再次计入。

4. 剩余期望用 $E = 1 + \sum pE_{next}$，吸收态取零。

> Example. Flip a fair coin until two consecutive heads appear. Find the expected number of flips.

> 例题：反复掷公平硬币，首次连续两次正面时停止，求掷币次数的期望。

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 7 / 方法 7 [CO05-M07] (continued / 续)

Worked solution

1. Let E_0 be the expectation with no terminal head and E_1 with one terminal head.

2. First-step equations are $E_0 = 1 + (E_0 + E_1)/2$ and $E_1 = 1 + E_0/2$; the second head reaches an absorbing state with zero remaining time.

3. Solving gives $E_0 = 6$. The time already spent in the next flip supplies the leading one in each equation.

例题解析

1. 令 E_0 为末尾无正面时的期望，E_1 为末尾恰有一个正面时的期望。

2. 首次步骤方程为 $E_0 = 1 + (E_0 + E_1)/2$、$E_1 = 1 + E_0/2$，再出正面进入剩余时间零的吸收态。

3. 解得 $E_0 = 6$。每式开头的一是下一次掷币已经花费的时间。

Merged states must have identical transition totals into every merged class, not merely look symmetric in a sketch.

合并状态须对每个合并类别具有相同转移总量，不能只凭图上看似对称。

For a prefix success boundary, retain only states with hit count $j \le \lfloor pn \rfloor$ at each required prefix length; a final total condition is an additional terminal filter. For predicting the next color of an unchanged deck, the prediction does not affect later information or transitions, so choosing the currently more numerous color maximizes the immediate success probability and is globally optimal.

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 7 / 方法 7 [CO05-M07] (continued / 续)

对前缀成功率边界，在每个要求的前缀长度只保留命中数 $j \le \lfloor pn \rfloor$ 的状态，最终命中总数另作终态筛选。预测剩余牌颜色时，预测本身不改变以后的信息与转移，因此每次猜当前数量较多的颜色最大化即时成功率，也给出整体最优策略。

Practice / 配套练习: AIME-2002-II-Q12 (WB p. 790), AIME-2018-I-Q14 (WB p. 684), AIME-2021-I-Q12 (WB p. 781)

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 8 / 方法 8 [CO05-M08]

Count exact covers with the first uncovered element

用首个未覆盖元素计算精确覆盖

Procedure

1. Represent every allowed piece by its set of occupied cells.

2. Choose the first uncovered cell in a fixed order and branch only over legal pieces containing it and disjoint from the covered set.

3. Stop with count one exactly when every cell is covered and the prescribed number of pieces is used.

4. Save the candidate list, recurrence, and resulting cover list or checksums so a computed answer can be independently reproduced.

通用步骤

1. 用占用单元集合表示每个允许部件。

2. 按固定顺序取首个未覆盖格，只对含该格且不与已覆盖集相交的部件分支。

3. 仅在全部单元覆盖且使用指定部件数时计一。

4. 保存候选表、递推与最终覆盖列表或校验值，使计算结果可独立复现。

> Example. Use three dominoes to cover a two-row, three-column board. Count the tilings by branching at the upper-left cell.

> 例题：用三块骨牌覆盖两行三列棋盘，按左上首格分支计算铺法。

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 8 / 方法 8 [CO05-M08] (continued / 续)

Worked solution

1. If its domino is vertical, the remaining two-by-two square has two covers: both vertical or both horizontal.

2. If it is horizontal, the lower-left cell must also belong to a horizontal domino; the last column is then vertical, giving one cover.

3. The branches are disjoint because the piece covering the chosen cell is unique. Total $2+1=3$.

例题解析

1. 若首格骨牌竖放，余下二乘二方块有全竖或全横两种。

2. 若首格横放，左下也须横放，末列被迫竖放，只有一种。

3. 各分支按覆盖首格的唯一部件区分，互不重复，共三种。

Pruning is valid only when its condition is proved necessary; the unpruned pivot recurrence already counts each partition once.

剪枝条件必须证明为必要条件；即使不剪枝，首格递推本身已保证每个分拆恰计一次。

In symbols, $C(U,j)=\sum_L C(U\cup L,j-1)$ over candidate pieces containing the first cell outside U and avoiding U. Set $C(U,0)=1$ for full coverage and zero otherwise. The 10×10 five-loop problem has a companion certificate listing all 2025 candidate boundary masks, all 83 accepted covers, and the exact search totals; no unproved geometric slab classification is needed for the count.

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符号上，$C(U,j)=\sum_L C(U\cup L,j-1)$，其中 L 含首个不在 U 中的单元且不交 U。$j=0$ 时仅全部覆盖取一。10×10 棋盘的五环题配有证书，记录全部二千零二十五个候选边界掩码、八十三个有效覆盖及精确搜索总量，计数不依赖未经证明的条块分类。

Practice / 配套练习: AIME-2026-I-Q15 (WB p. 688)